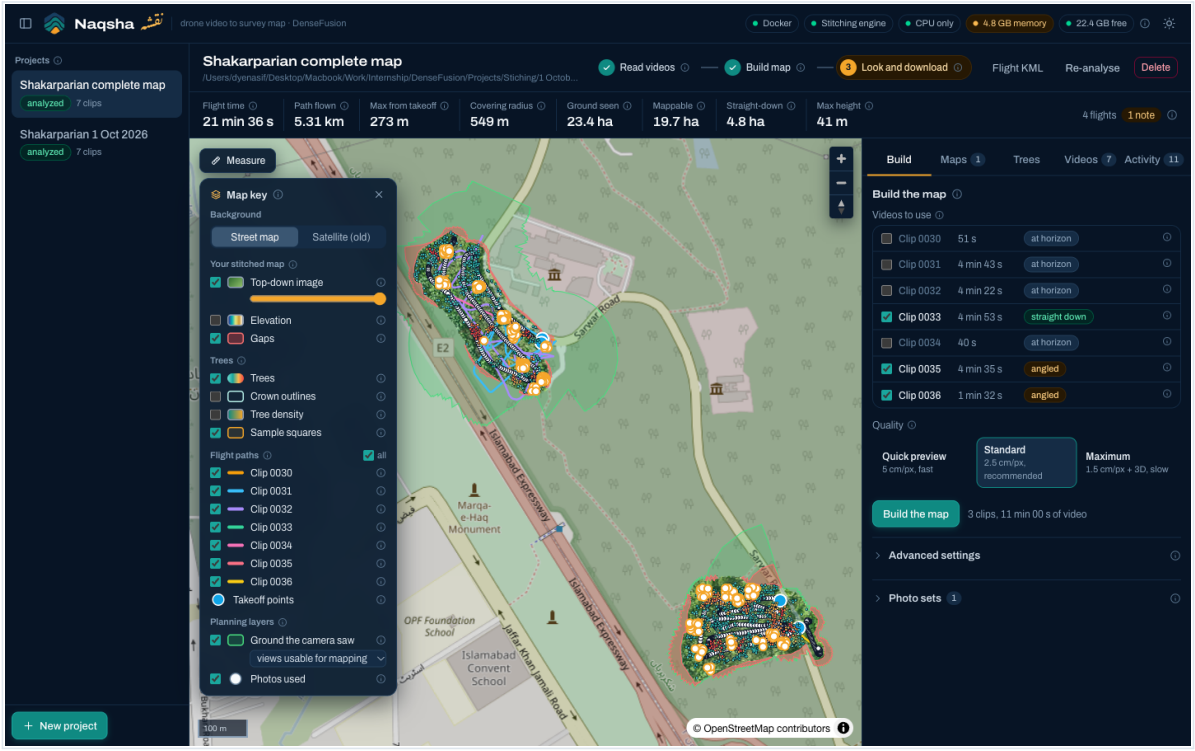
Naqsha نقشہ

User Guide

Every screen, every button, and where your files go: from a folder of drone videos to a finished map and a tree count.

DENSEFUSION · DRONE VIDEO TO SURVEY MAP · OCTOBER 2026



The main screen after a map has been built: the flight summary at the top, the map in the middle with its key on the left, and the Build / Maps / Videos / Activity panel on the right. Every small (i) explains the thing next to it when you hover over it.

In one sentence: point Naqsha at a folder of drone videos, press Build the map, wait, and download a stitched top-down map with coordinates, an elevation map and the flight paths. Everything in between is automatic.

Contents. 1 Starting the app. 2 Creating a project. 3 Reading the flight summary. 4 The map and its key. 5 Looking at individual photos. 6 Building a map. 7 Watching progress. 8 The Maps tab: quality and downloads. 9 Exploring the result. 10 The Videos tab. 11 Counting trees. 12 Where your files are, and changing a map. 13 Running without the browser. 14 Troubleshooting. Tip: anywhere in the app, hover over a small (i) for a plain explanation of what that part does.

1. Starting the app

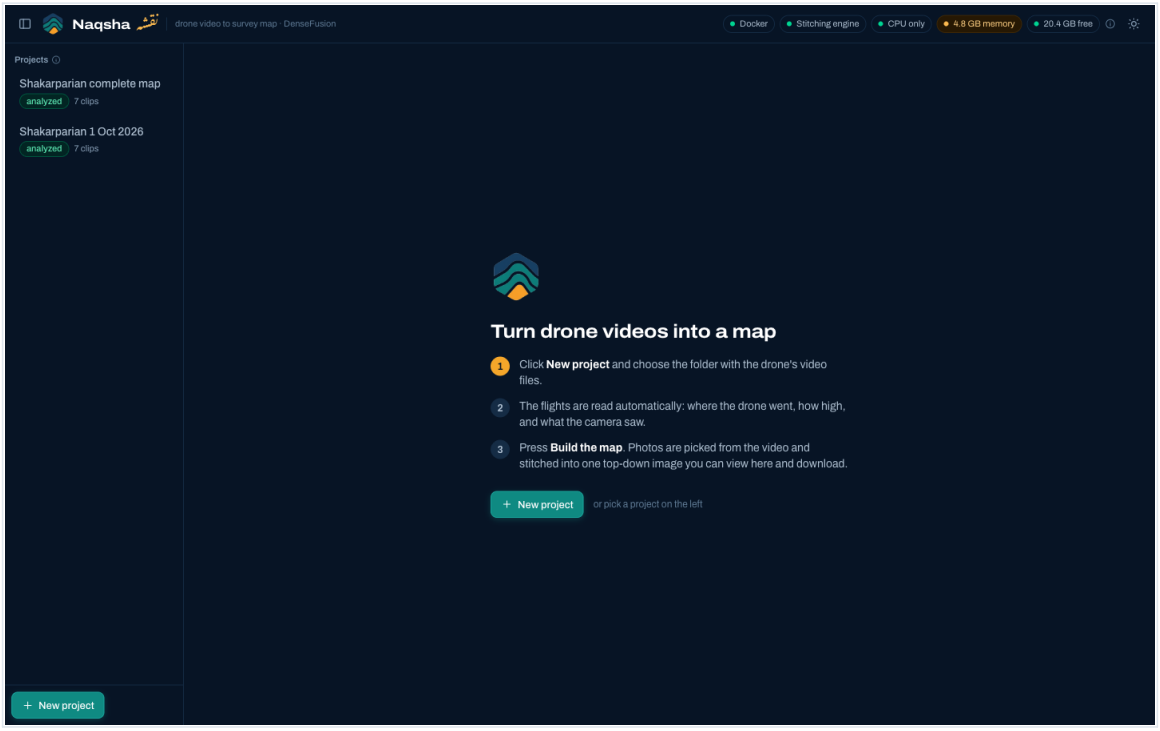
Naqsha runs on your own computer. Nothing is uploaded anywhere.

- Start Docker Desktop first (the whale icon in the menu bar). The stitching engine runs inside it.
- Open Terminal, go to the `naqsha` folder and run `./run.sh`. Leave this window open while you work.
- When you click a button that starts work, it shows a small spinning circle and cannot be clicked again until the work has started, so there is no need to click twice.
- Open http://127.0.0.1:8765 in your browser (Safari or Chrome).

```
cd ~/Desktop/Macbook/Work/Internship/DenseFusion/Projects/Stiching/naqsha
./run.sh
```

The small labels at the top right show whether the computer is ready: Docker, Stitching engine (installed), CPU only or GPU, how much memory the stitching engine may use, and free disk space; the (i) at the end explains each. A label turns saffron when something deserves attention, for example less than 20 GB of free disk. If an NVIDIA graphics card is available (on the GPU computer), Naqsha uses it automatically; on a Mac, stitching uses the processor.

The sun and moon button at the far top right switches between dark and light mode; the app remembers your choice.



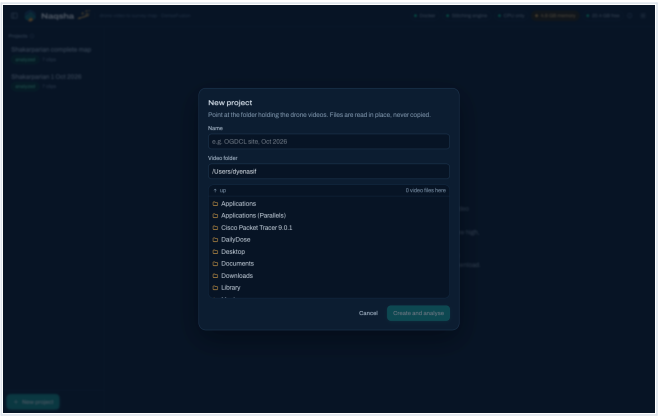
The start screen. Your projects are listed on the left; the button at the top left hides or shows that list.

2. Creating a project

Click New project. Give it a name, then pick the folder that holds the drone's video files: click folder names to go into them and up to go back. The count at the top right tells you how many videos are in the folder you are looking at.

Click Create and analyse. The videos are read where they are; nothing is copied.

Within about a minute Naqsha reads the GPS, height and camera angle stored inside every video and draws the flights on the map. This is step 1 of the three steps shown next to the project name; the current step is highlighted in saffron and finished steps get a tick.



Choosing the video folder.

3. Reading the flight summary

The strip of numbers under the project name answers "where did the drone fly?". Hover over the (i) next to any number for a short explanation.

Number	What it means
Flight time	total length of all videos
Path flown	distance the drone travelled, from its cleaned GPS track
Max from takeoff	the farthest the drone got from where it took off
Covering radius	radius of the smallest circle that contains the whole flight (its centre is in the tooltip)
Ground seen	all the ground any camera frame showed
Mappable	ground seen with the camera at least 30 degrees below the horizon (usable for a map)
Straight-down	ground seen with the camera pointing almost straight down (best quality)
Max height	highest point above the ground, using a terrain model

At the right end of the strip, the (i) beside "4 flights" shows each flight's takeoff point and any notes, for example that the drone's own altitude readings disagreed between batteries and were corrected.

4. The map and its key

The Map key panel on the left of the map names every colour and line and lets you switch each one on or off. The (i) on each row explains it.

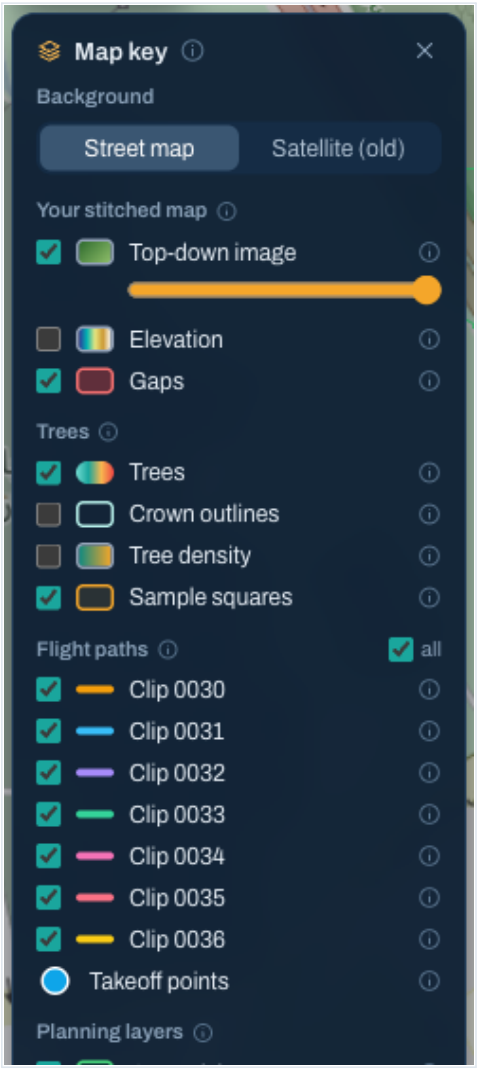
Background: a street map, or old satellite imagery for comparison (several years old).

Your stitched map: the finished map, with a slider to fade it in and out; the elevation colours; and gaps (red) where the drone flew but nothing could be stitched.

Flight paths: one colour per video, with whether its camera angle is useful for mapping. Untick a clip to hide its line.

Planning layers: the ground the camera saw (green), and the photos used for the map (white dots).

The cross closes the key; the Map key button brings it back.



Hover to find out what anything is. Moving the mouse over a flight path, takeoff point, coverage area or gap shows a short explanation.

Zoom with the mouse wheel or the + and - buttons; drag to move around.



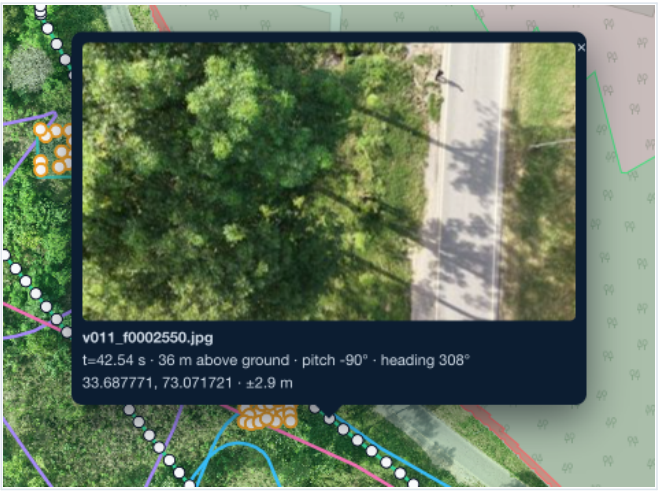
Hovering over a flight path says which video and flight it is.

5. Looking at individual photos

Each white dot is one video frame that was picked as a photo for the map. Click a dot to see the photo, when it was taken in the video, its height above the ground, camera angle, heading and position accuracy.

Click the photo to open it full size.

After a map is built, dots turn red if the photo landed far from its GPS position (it is left out automatically) and purple if it could not be fitted in.



A photo picked from the video, with where and how it was taken.

6. Building a map

Open the Build tab (step 2).

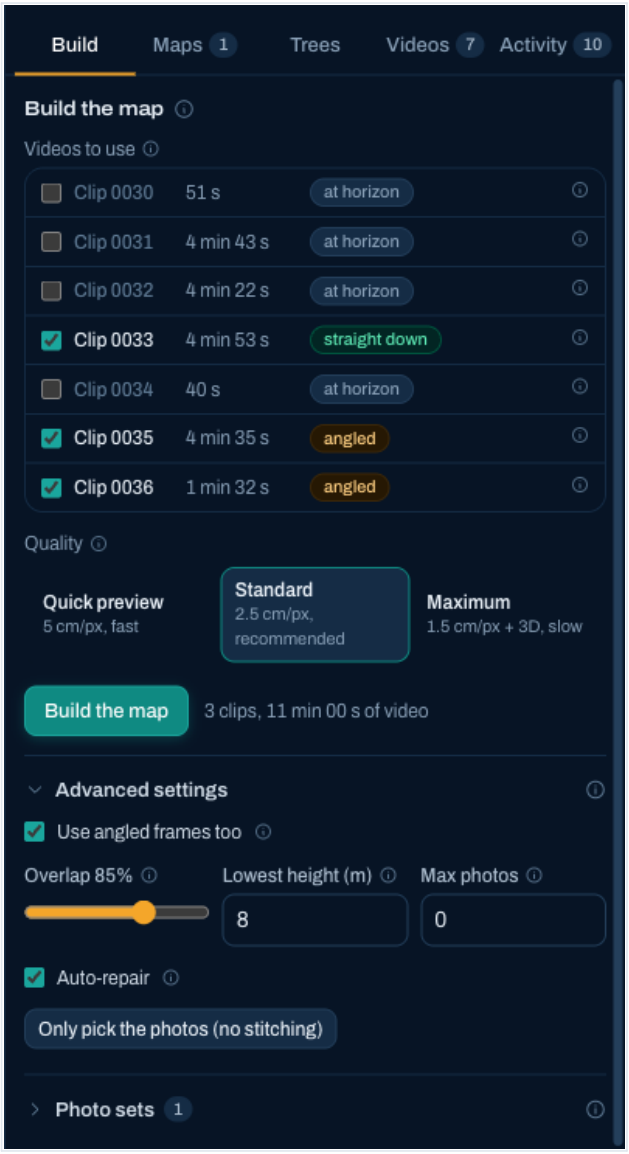
Pick the clips. Naqsha ticks the useful ones for you. Clips marked at horizon filmed the horizon and add little; angled clips are usable; straight down is best.

Pick a quality. Quick preview takes minutes and is good for checking coverage. Standard (recommended) gives about 2.5 cm per pixel plus an elevation map. Maximum detail is meant for the GPU computer.

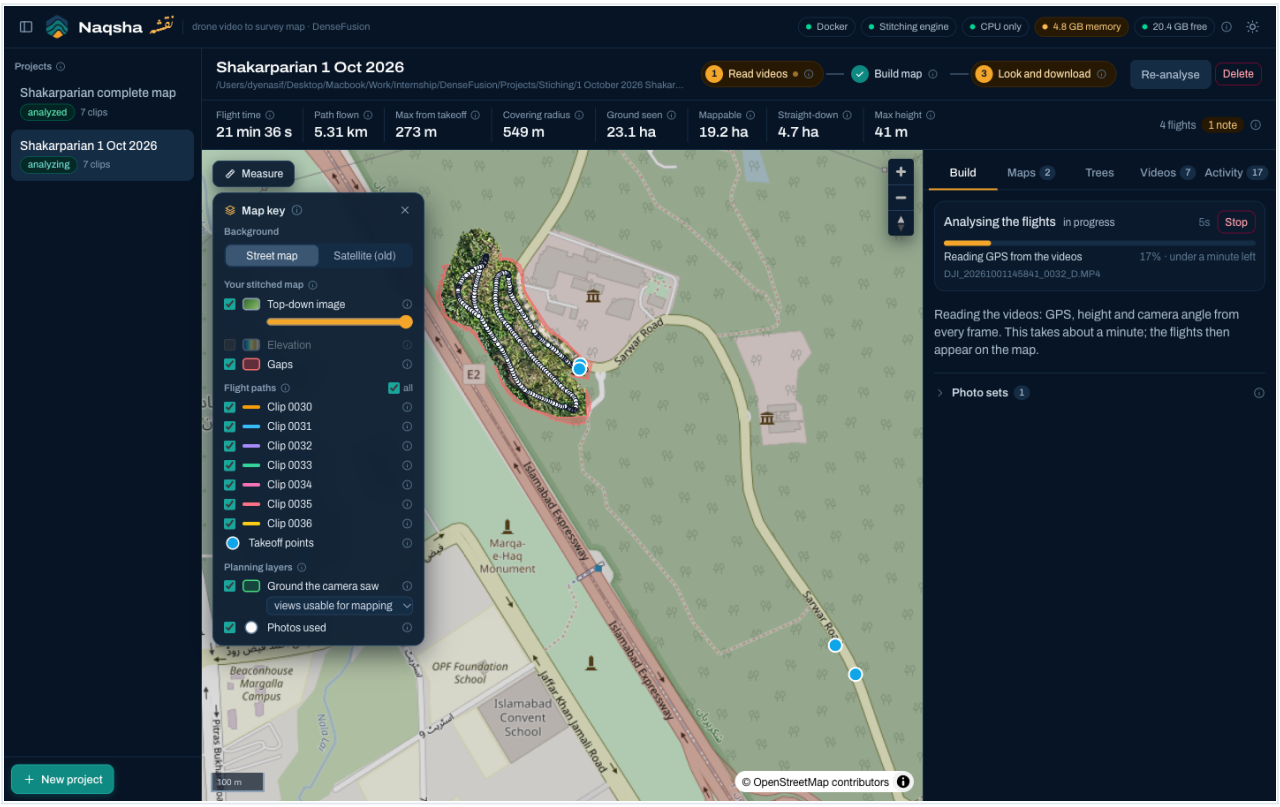
Press Build the map.

Advanced settings (optional, folded away until you open it): use angled frames; overlap between neighbouring photos (85% is right for trees, more is safer but slower); the lowest height to use; a cap on the number of photos; and auto-repair, which rebuilds once if the check finds misplaced photos. Only pick the photos prepares a photo set without stitching, so you can inspect it first.

Large surveys are processed in parts that fit in this computer's memory and joined at the end. You do not need to do anything about this.



7. Watching progress



A running job: what it is doing in plain words, the percentage and an estimate of the time left.

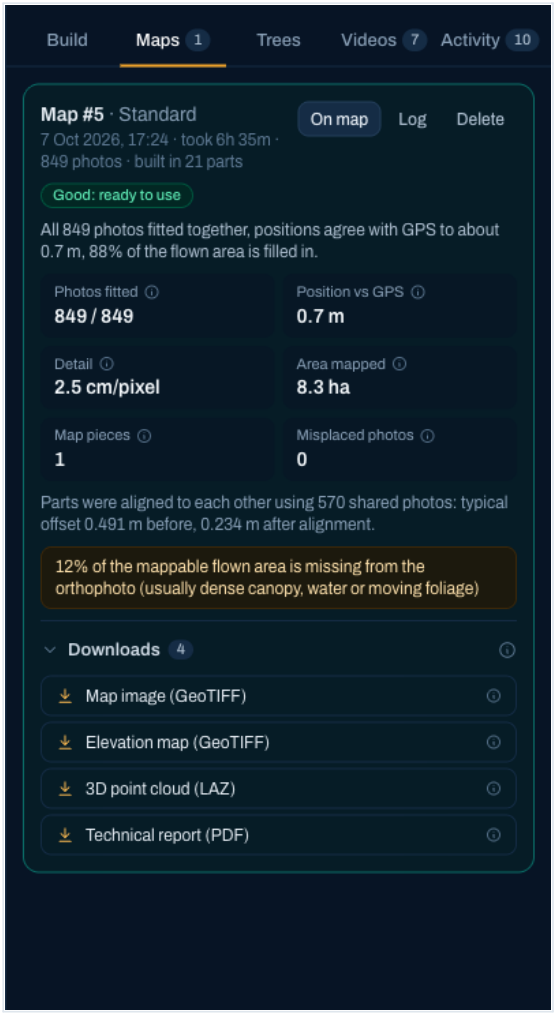
- A progress card at the top of the right-hand panel shows the current stage, for example "Matching photos to each other" or "Part 4 of 17: Building the dense 3D surface", with an estimated time left.
- Stop cancels the job. A stopped or failed map keeps the parts that already finished, and Retry on the map card continues from there.
- You can close the browser; the work continues as long as the Terminal window running `./run.sh` stays open.
- Timing on an 8 GB Mac: reading the videos about a minute; a quick preview of 300 photos about 10 minutes; a standard map of 850 photos several hours. The GPU computer is much faster.
- If the computer runs short of memory, the affected part is split and redone automatically. You will see the plan grow (for example from 15 to 21 parts); this is normal.

8. The Maps tab: quality and downloads

Every map gets a verdict in plain words (Good: ready to use, Usable, with some gaps, or Poor) followed by one sentence that sums it up, then the numbers below. Each number has an (i) explaining it.

On map / Show on map displays that map; Log shows the engine's technical log; Delete removes the map (the videos are never touched).

Downloads at the bottom of the card: the map image, the elevation map, a 3D point cloud and a technical report. The (i) on each says what the file is and which programs open it.



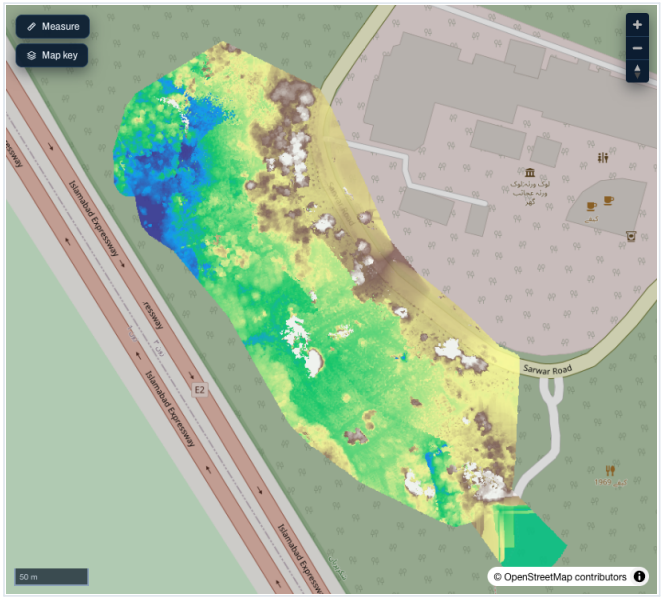
A finished map: verdict, numbers and downloads.

Item	What it tells you
Photos fitted	how many photos the engine could place; ideally all
Position vs GPS	typical difference between where the map puts each photo and the drone's GPS
Detail	the size of one map pixel on the ground
Area mapped	how much ground the map covers, and what share of the flown area that is
Map pieces	1 means everything joined into one consistent map
Misplaced photos	photos that landed far from their GPS (left out by auto-repair)
Parts aligned	for big maps: how far apart the parts were before and after joining

Past jobs (finished, stopped or failed) are listed in the Activity tab.

9. Exploring the result





Left: the stitched map close up (2.5 cm per pixel). Right: the elevation colours, where trees stand out from the ground.





Left: the new map over the old satellite background, with red gaps at its edges. Right: flight paths, coverage (green) and gaps (red) over the street map.

Measuring. Click Measure (top left of the map), then click points on the map. The box shows the total distance, and the enclosed area once you have three or more points. Click Stop measuring to finish.

Use the elevation layer and the fade slider together to compare the map with the background.



Measuring a boundary: distance and enclosed area.

10. The Videos tab

One row per video: its length, typical height above the ground and how much of it looked straight down. Click a row for the rest: distance flown, which flight it belonged to, camera angle and GPS quality.

GPS jumps (sudden steps in the drone's position) are shown in saffron. They are smoothed automatically.

The badge on each row says whether the clip is useful for a map.



11. Counting trees

Open the Trees tab on a finished map and press Count trees (about two minutes; five with a model taught on your site). Naqsha finds every tree crown in the map image with DeepForest, a deep learning model trained on hand-labelled crowns, and measures each tree's height from the elevation map.

Taller than picks what counts as a tree: All includes everything woody from 1 m (young trees and bushes); 4 m counts established trees only. The headline, the map dots and the downloads all follow it, and it is saved for the map. By height shows how many fall in each band, so you can read any definition.

Not yet checked means the number has not been compared with a hand count. Mark every tree in a few random test squares and Naqsha measures its own accuracy and gives a range.



The Trees tab: count, height filter and height bands.

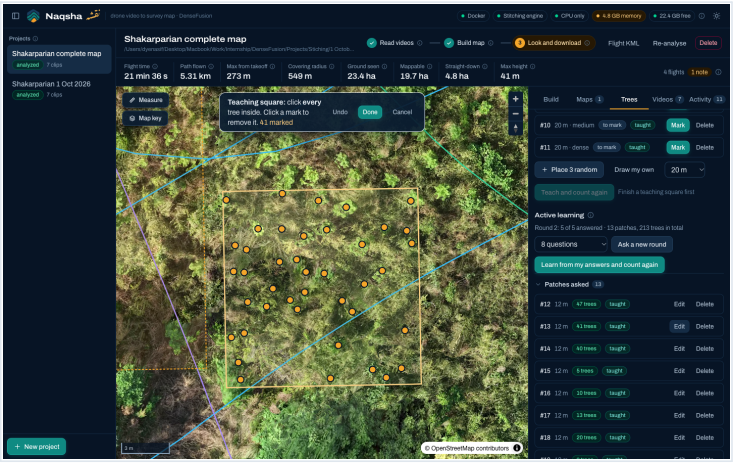
Each dot is one tree, coloured by height (teal short, saffron tall). Hover a dot for its height, crown width and how it was found. In the map key, Crown outlines and Tree density (trees per hectare in 20 m squares) can be switched on.



Trees on the map, at least 4 m tall.

Teaching the model your trees. The image model was trained elsewhere, so it improves a lot when it sees this site. In Active learning press Ask me: Naqsha places small 12 m patches where it is least sure and takes you to each in turn. Click every tree inside, then Next; a patch with no trees is a real answer.

Your own trees are hidden while you mark, so they cannot steer you. Press Learn from my answers and count again (about five minutes); repeat with Ask a new round for further improvement.



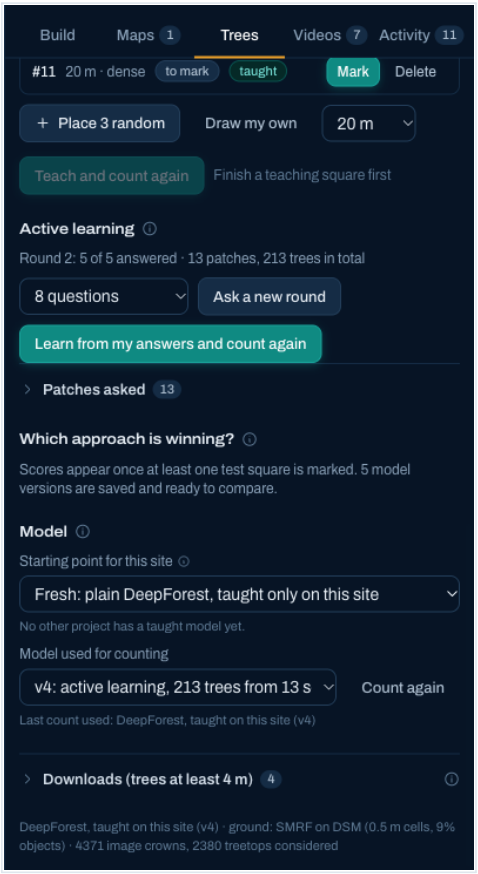
Marking a patch: click every tree inside the square.

Test squares (saffron) are placed at random and only grade the count. Teaching squares and active-learning patches (teal) only teach. Keeping them apart is what makes the accuracy honest.

Teaching always learns from everything marked on this site: random teaching squares and active-learning answers together, so both approaches improve one model.

Starting point: every new project starts from the plain DeepForest model, so another site never steers it. If a site with similar vegetation was already taught, you can start from that model instead. Model lists this site's taught versions; switch and press Count again.

Mark young trees and bushes the same way everywhere: whatever you mark is what the model learns to count. The height filter then decides what you report.



Teaching, active learning and model versions.

12. Where your files are, and changing a map

The finished Shakarparian map is in the Shakarparian Map (Naqsha output) shortcut on your Desktop, which opens this folder:

```
Desktop/Macbook/Work/Internship/DenseFusion/Projects/Stiching/
    1 October 2026 Shakarparia scan/naqsha-output/
```

File	What it is	Open it with
map.tif	the stitched map with coordinates (427 MB)	QGIS (free), ArcGIS, Google Earth Pro, Global Mapper
map_preview.jpg	a quick-look picture of the whole map	any image viewer
elevation.tif	surface height for every pixel	QGIS, ArcGIS
flight.kml	flight paths and takeoff points	Google Earth
coverage.geojson, track.geojson	covered ground and tracks	QGIS, geojson.io
quality.json	the full quality report	any text editor

Everything else Naqsha keeps (projects, extracted photos, parts of maps) lives in `Stiching/naqsha-data/`. Delete projects from the app rather than by hand.

Can I change the map? Yes, by building it again with different choices: other clips, another quality, more overlap, or angled frames on or off. Each build becomes a new entry in the Maps tab, so you can compare them and delete the ones you do not need. A photo set can also be re-stitched at a different quality with Make a map from these photos, without picking photos again. Naqsha does not edit map pixels by hand; for drawing, cutting or annotating, open `map.tif` in QGIS. For better absolute accuracy (survey grade), a few ground control points measured on site are needed; that is a planned addition.

13. Running without the browser

One command does everything, from reading the videos to exporting the files, without any clicking:

```
./run.sh process "/path/to/video folder"
./run.sh process "/path/to/video folder" --quality preview --out ~/Desktop/my-map
./run.sh process "/path/to/video folder" --trees --tree-min-height 4
```

It prints progress as it goes and writes the results to `naqsha-output` next to the videos (or to the folder given with `--out`). The project also appears in the web app, so you can look at it there afterwards.

14. Troubleshooting

What you see	What to do
Docker off label, or "Docker is not running"	start Docker Desktop and wait until its whale icon is steady
Download stitching engine label	click it once; the stitching engine (about 2 GB) downloads in the background
A map failed	open the Maps tab, read the first line of the error, and press Retry; finished parts are kept
"ran out of memory even after splitting"	close other apps, or choose a lower quality; on the Mac keep Docker at 5 GB
The Mac feels slow during a build	normal: stitching uses all processor cores. Light use (browsing, video) is fine
Disk label turns saffron	free space, or delete old maps and photo sets in the app
A clip says "no telemetry"	that video has no GPS stored in it and cannot be placed on the map
Big red gaps over trees	dense canopy is hard to stitch; fly higher or with more overlap next time